

# Amazon Product Analysis

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# Abstract

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Amazon is a multinational e-commerce company and has experienced a 41% decline in product purchases during its Prime Day event in 2025 compared to the previous year. As a result, both the marketing and sales departments at Amazon are looking to improve their customer experience. They seek the help of the Lead Data Analyst to provide them with a comprehensive product report which will examine the products carried by Amazon. The marketing department will use this information to deliver product-specific digital promotion while the sales department will use this information to suggest a range of products most suitable to each customer.

## Research Questions

A product analysis will be performed for the marketing and sales departments with a focus on answering the following research questions.

### Question 1

How does the price, rating and discount compare for products belonging to different categories?

#### **Purpose**

Obtain an overview of the products sold by Amazon to make it easier to implement guided marketing and sales strategies.

#### **Impact Hypothesis**

When a sales employee is making efforts to sell a product to a future customer, the employee will have sufficient knowledge to answer any questions about the product, including questions relating to pricing and potential discounts.

### Question 2

How can the product characteristics be used to predict the product category?

#### **Purpose**

Classify new products to ensure appropriate marketing and sales strategies are applied for market success.

#### **Impact Hypothesis**

A member of the *Marketing* team is asked to curate digital content to promote a new product. In this case product category classifier models can be used to identify the already existing digital materials prepared for products belonging to the same category.

### Question 3

How can product descriptions be used to identify similar products?

#### **Purpose**

Match new products to existing products for segmentation and recommendation strategies.

**Impact Hypothesis**

A sales employee needs to find products similar to the one described by a customer using the product information retrieval models.

## Methodology

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### Preliminary Analysis

Extract descriptive statistics such as means and frequencies. Compute correlations to observe potential relationships between variables. Visualize variable distributions to observe general trends and outliers using boxplots and histograms.

### Classification Models

Using the results of the preliminary analysis, construct three different product category classifiers, a linear multinomial logistic regression model and two nonlinear classifiers: a k-nearest neighbours model and a decision tree model. Compare the three models using evaluation metrics such as accuracy, precision and recall.

### Information Retrieval Models

Construct multiple unigram and bigram vector space models to process product descriptions. Apply different text preprocessing techniques (stopword removal, lemmatization, etc.) and use the TD-IDF statistical weight measure to evaluate importance of words. Calculate retrieval accuracy by category to evaluate search relevance.

## Literature Review

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### Predicting Purchase Intention Categories Using Logistic Regression

E-commerce operations expand beyond the online store itself, with advertisements for products spreading across various social media platforms. While data from such platforms can provide key insights on user behaviour, it can be difficult to accurately analyze the hidden effects of the inherent time-series data (Wu, 2025). Wu proposes the use of machine learning algorithms and their abilities to capture complex patterns in the data to predict the category of users' purchase intentions. These categories include those which are common to many large e-commerce retailers today such as beauty, digital, food, clothing, etc. While structured user data can be

used to build and train models like logistic regression, the lack of temporal and textual data can hinder prediction accuracy and marketing strategies (Wu, 2026).

Wu proposes the use of a Transformer-BiGRU classification algorithm for its ability to handle temporal dependencies in behavioural data unlike traditional models. The study found that the proposed predictive model outperformed all other models when comparing evaluation metrics. Indeed, as suggested prior by Wu, the logistic regression model delivered nearly the lowest performance for accuracy, precision and recall. Notably, while the decision tree and KNN models were not as effective as the Transformer-BiGRU, they still recorded higher metrics than the logistic regression model.

This study validates the use of a multinomial logistic regression model in the case of predicting retail categories in an e-commerce setting. Although this study is focused on using user data to predict the category of the product intended to be purchased, our second research question of interest will instead require product characteristics to be used to predict the corresponding product category instead. Additionally, a key finding is the possible reduction in prediction accuracy when excluding textual data such as product descriptions, in the logistic regression model.

## **Usage of KNN in E-Commerce Recommendation Systems**

Recommendation systems are the core of all e-commerce systems today. These systems deploy algorithms which are designed to create a unique and personalized online shopping experience for all users. They work by filtering information from a variety of user-based data sources including purchase and browsing history. (Kong et al., 2024). In this study, a k-nearest neighbours (KNN) classification model is used to construct the recommendation system. Compared to other models, it has the shortest training time but with poor performance. This can be attributed to the model's inability to capture complex patterns between users and their purchases (Kong et al., 2024).

In the case of our study, there is no data which describes the relationship between users and products besides user reviews, which will not be considered for our analysis. Our analysis relies solely on using product characteristics to predict the product category and as a result, the limitation addressed by Kong does not apply.

## **Influence of Product Characteristics on Online Preferences**

The evolution of e-commerce systems has created the need to understand the factors which have the most influence in shaping the online shopping preferences of consumers. Such research can help a business determine the viability of an e-commerce model. Frequent users of e-commerce platforms were asked to complete a survey which was designed to collect data on the various characteristics of online products and how these related to their desire to shop online. Lee uses a binary decision tree model to predict the preference to shop online as

supposed to in-person. The purchase frequency and price showed to have the highest influence on the decision to shop online (Lee et al., 2007). In addition, the decision tree model offered higher performance accuracy compared to logistic regression.

A key difference between this study and ours is the use of a decision tree model for binary versus nonbinary classification. This study however, enforces the use of such a model with similar product input features to analyze e-commerce sales data.

## **Classifying Product Descriptions Using TF-IDF**

Term frequency-inverse document frequency (TF-IDF) is a statistical weight measuring technique which is used to both measure and rank important words from a textual document (Dagobert et al., 2023). In this study, the output of such an algorithm is fed into a support vector machine (SVM) classification system to predict the product category given a product description. Feature extraction for the support vector machine (SVM) algorithm is completed by constructing a TF-IDF matrix after applying tokenization and stopword removal preprocessing techniques to e-commerce product descriptions. The SVM reported high accuracy highlighting the effectiveness of the TF-IDF algorithm in detecting relevant terms in product descriptions.

In contrast, our third research question focuses on using product descriptions to identify similar products. This study enforces the use of the TF-IDF statistical method to evaluate the importance of words in product descriptions, allowing for us to proceed with this method to identify similar products using an IR model.

## **Identifying Customer Trends Using EDA**

E-commerce platforms generate millions of transactions daily which contain valuable information about the ways consumers interact with the different products available for purchase. Conducting an exploratory data analysis (EDA) can help to identify relationships between consumer behaviour and product sales (Chakma et al., 2025). The results of such an analysis can be useful for advanced statistical manipulation including collaborative filtering models for recommendation systems. Chakma observes statistical summaries of the data such as variable counts, means, and percentiles in addition to creating bar plots to visualize customer age distributions. Simple mathematical computations are also carried out, such as average purchase cost by age group.

In another study focused on understanding retail patterns for Amazon India, heat maps are constructed to visualize sales by factoring in variables like state, category and garment size. Time-based analysis is also done through line graphs to analyze sales trends by month and day. These high-level analytics revealed key insights related to consumer behaviour, such as Maharashtra being the state with both the highest revenue and consumer activity when compared to others (Ali et al., 2026).

While our first research question does not require the use of any temporal data, we will still be using product data to compare differences among the various product categories. These studies have demonstrated the effectiveness of utilizing EDA methods in the context of an e-commerce setting to highlight key information prior to further statistical analyses.

## **Integration and Overview of Findings**

We developed three models to classify products by category out of which the decision tree reported the best performance followed by the logistic regression and KNN models. Chakma's study supports the construction of various forms of descriptive summaries of product data which we followed prior to the modeling phase of our project.

Similar to Wu's study, we also used the models to classify product categories with a key difference being the usage of product characteristics as opposed to user data for the model features. Our KNN model reported the lowest scores of accuracy, precision and recall while for Wu, it was the logistic regression model. In our case, logistic regression only slightly outperformed the KNN classifier and Wu suggests that this lack in performance could be due to the inability of standard machine learning models to handle textual data.

It should also be noted that after rigorous testing to validate the assumptions of logistic regression, our model was developed using one less feature compared to the KNN and decision tree models which could possibly explain the differences in final model performances when compared to Wu's research.

When comparing our usage of the KNN model to predict product categories to its application in e-commerce recommendation systems, we also noted a short training time compared to other models with poor performance as previously highlighted.

Predicting consumer preferences to shop online versus in-person was discussed using the approach of a binary decision tree with prices being a highly influential factor. Similarly, in our study we found product prices to be an important feature when using a multiclass decision tree model for product category classification. In both cases, the decision trees outperformed the logistic regression models when comparing performance accuracy.

Constructing TF-IDF matrices using product descriptions was shown to be a strong method of feature extraction for the SVM algorithm. Likewise, using these matrices for the development of vector space models to retrieve similar products using product descriptions also displayed high levels of accuracy.



## Data Analysis

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The *Amazon Sales Dataset* will be the primary source of data for this project which contains over 1000 authentic Amazon product records scraped using Python. The dataset contains 16 variables, and *Figure 1* provides an overview of the relevant characteristics of the dataset.

Figure 1: Overview of Variables

| Name                | Description                                          | Type                 |
|---------------------|------------------------------------------------------|----------------------|
| product_name        | String containing product name                       | Text                 |
| category            | Pipe separated string containing product categories  | Text                 |
| discounted_price    | Discounted price of product in Indian Rupees (INR)   | Numerical with units |
| actual_price        | Price of product in INR                              | Numerical with units |
| discount_percentage | Percentage discount on product                       | Numerical with units |
| rating              | Average rating of product from ranging from 1-5      | Numerical            |
| rating_count        | Number of individuals who voted for a product rating | Numerical            |
| about_product       | Description of product                               | Text                 |

Figure 2: Overview of Data

|   | product_id | product_name                                      | category                                          | discounted_price | actual_price | discount_percentage | rating | rating_count | about_product                                      |
|---|------------|---------------------------------------------------|---------------------------------------------------|------------------|--------------|---------------------|--------|--------------|----------------------------------------------------|
| 0 | B07JW9H4J1 | Wayona Nylon Braided USB to Lightning Fast Cha... | Computers&Accessories Accessories&Peripherals ... | ₹399             | ₹1,099       | 64%                 | 4.2    | 24,269       | High Compatibility : Compatible With iPhone 12...  |
| 1 | B098NS6PVG | Ambrane Unbreakable 60W / 3A Fast Charging 1.5... | Computers&Accessories Accessories&Peripherals ... | ₹199             | ₹349         | 43%                 | 4.0    | 43,994       | Compatible with all Type C enabled devices, be...  |
| 2 | B096MSW6CT | Sounce Fast Phone Charging Cable & Data Sync U... | Computers&Accessories Accessories&Peripherals ... | ₹199             | ₹1,899       | 90%                 | 3.9    | 7,928        | 【 Fast Charger& Data Sync】 -With built-in safet... |
| 3 | B08HDJ86NZ | boAt Deuce USB 300 2 in 1 Type-C & Micro USB S... | Computers&Accessories Accessories&Peripherals ... | ₹329             | ₹699         | 53%                 | 4.2    | 94,363       | The boAt Deuce USB 300 2 in 1 cable is compati...  |
| 4 | B08CF3B7N1 | Portronics Konnect L 1.2M Fast Charging 3A 8 P... | Computers&Accessories Accessories&Peripherals ... | ₹154             | ₹399         | 61%                 | 4.2    | 16,905       | [CHARGE & SYNC FUNCTION]- This cable comes wit...  |

Figure 3: Missing Values

|                     |   |
|---------------------|---|
| product_id          | 0 |
| product_name        | 0 |
| category            | 0 |
| discounted_price    | 0 |
| actual_price        | 0 |
| discount_percentage | 0 |
| rating              | 0 |
| rating_count        | 2 |
| about_product       | 0 |

The rating count for two products in the dataset has not been provided.

## Data Cleaning

The missing values will be removed as these account for only a small proportion of our data. Characters and punctuation will be removed from the numerical variables before converting them to their float equivalents. Finally, any duplicate product records will be removed to reduce model bias.

Figure 4: Cleaned Dataset

|   | product_id | product_name                                      | category                                          | discounted_price | actual_price | discount_percentage | rating | rating_count | about_product                                      |
|---|------------|---------------------------------------------------|---------------------------------------------------|------------------|--------------|---------------------|--------|--------------|----------------------------------------------------|
| 0 | B07JW9H4J1 | Wayona Nylon Braided USB to Lightning Fast Cha... | Computers&Accessories Accessories&Peripherals ... | 399.0            | 1099.0       | 64.0                | 4.2    | 24269.0      | High Compatibility : Compatible With iPhone 12...  |
| 1 | B098NS6PVG | Ambrane Unbreakable 60W / 3A Fast Charging 1.5... | Computers&Accessories Accessories&Peripherals ... | 199.0            | 349.0        | 43.0                | 4.0    | 43994.0      | Compatible with all Type C enabled devices, be...  |
| 2 | B096MSW6CT | Sounce Fast Phone Charging Cable & Data Sync U... | Computers&Accessories Accessories&Peripherals ... | 199.0            | 1899.0       | 90.0                | 3.9    | 7928.0       | 【 Fast Charger& Data Sync】 -With built-in safet... |
| 3 | B08HJD86NZ | boAt Deuce USB 300 2 in 1 Type-C & Micro USB S... | Computers&Accessories Accessories&Peripherals ... | 329.0            | 699.0        | 53.0                | 4.2    | 94363.0      | The boAt Deuce USB 300 2 in 1 cable is compati...  |
| 4 | B08CF3B7N1 | Portronics Konnect L 1.2M Fast Charging 3A 8 P... | Computers&Accessories Accessories&Peripherals ... | 154.0            | 399.0        | 61.0                | 4.2    | 16905.0      | [CHARGE & SYNC FUNCTION]- This cable comes wit...  |

## Summary of Variables

Figure 5: Summary of Quantitative Variables

|       | actual_price  | discounted_price | discount_percentage | rating_count  | rating      |
|-------|---------------|------------------|---------------------|---------------|-------------|
| count | 1349.000000   | 1349.000000      | 1349.000000         | 1349.000000   | 1349.000000 |
| mean  | 5698.133150   | 3309.369288      | 46.639733           | 17644.494440  | 4.089696    |
| std   | 11225.528577  | 7178.313092      | 21.615828           | 42145.645569  | 0.306806    |
| min   | 39.000000     | 39.000000        | 0.000000            | 2.000000      | 1.000000    |
| 25%   | 899.000000    | 349.000000       | 31.000000           | 1106.000000   | 3.900000    |
| 50%   | 1795.000000   | 899.000000       | 49.000000           | 4740.000000   | 4.100000    |
| 75%   | 4590.000000   | 2179.000000      | 62.000000           | 16020.000000  | 4.300000    |
| max   | 139900.000000 | 77990.000000     | 94.000000           | 426973.000000 | 5.000000    |

The price of products ranges from ₹39 to ₹139,900 with the average price being about ₹5,698. This suggests that there are extreme outliers present in both the actual and discounted price variables.

Figure 6: Overview of Product Categories

| category                                                                                                                | count |
|-------------------------------------------------------------------------------------------------------------------------|-------|
| Computers&Accessories Accessories&Peripherals Cables&Accessories Cables USBCables                                       | 233   |
| Electronics WearableTechnology SmartWatches                                                                             | 76    |
| Electronics Mobiles&Accessories Smartphones&BasicMobiles Smartphones                                                    | 68    |
| Electronics HomeTheater,TV&Video Televisions SmartTelevisions                                                           | 63    |
| Electronics Headphones,Earbuds&Accessories Headphones In-Ear                                                            | 52    |
| ...                                                                                                                     | ...   |
| Home&Kitchen Kitchen&HomeAppliances SmallKitchenAppliances RotiMakers                                                   | 1     |
| Home&Kitchen Heating,Cooling&AirQuality Parts&Accessories FanParts&Accessories                                          | 1     |
| Home&Kitchen Kitchen&HomeAppliances SmallKitchenAppliances StandMixers                                                  | 1     |
| Home&Kitchen Heating,Cooling&AirQuality Fans PedestalFans                                                               | 1     |
| Home&Kitchen Kitchen&HomeAppliances Vacuum,Cleaning&Ironing Vacuums&FloorCare VacuumAccessories VacuumBags HandheldBags | 1     |

211 rows x 1 columns

There are 211 product categories 205 of which correspond to only one product. To address this massive imbalance, only the categories with 30 or more products will be used for the analysis.

Figure 7: Product Categories with 30 or More Products

| category                                                                          | count |
|-----------------------------------------------------------------------------------|-------|
| Computers&Accessories Accessories&Peripherals Cables&Accessories Cables USBCables | 159   |
| Electronics Mobiles&Accessories Smartphones&BasicMobiles Smartphones              | 68    |
| Electronics WearableTechnology SmartWatches                                       | 62    |
| Electronics HomeTheater,TV&Video Televisions SmartTelevisions                     | 60    |
| Electronics Headphones,Earbuds&Accessories Headphones In-Ear                      | 51    |
| Electronics HomeTheater,TV&Video Accessories RemoteControls                       | 49    |

## Measures of Central Tendency

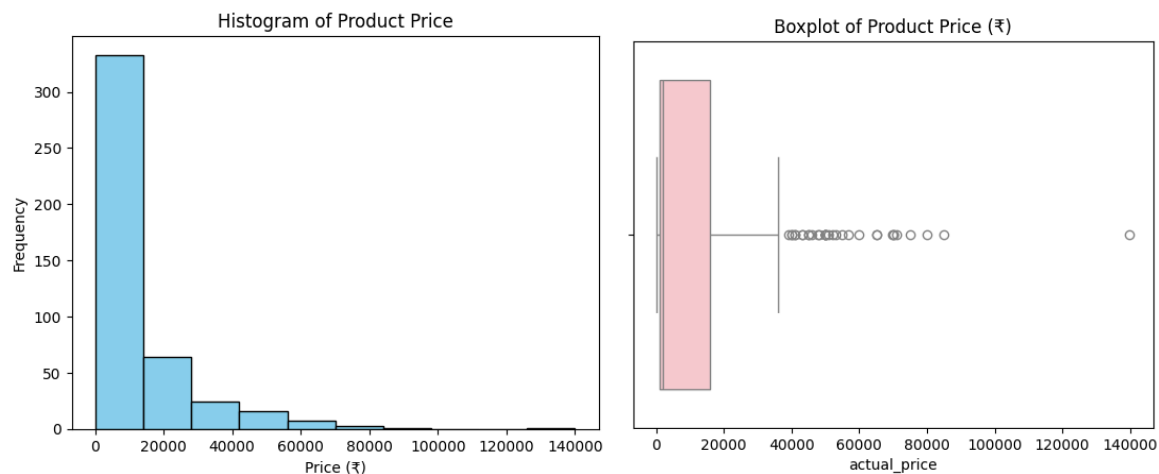
Figure 8: Averages by Product Category

|                                                                                   | actual_price | discounted_price | discount_percentage | rating_count | rating   |
|-----------------------------------------------------------------------------------|--------------|------------------|---------------------|--------------|----------|
| category                                                                          |              |                  |                     |              |          |
| Computers&Accessories Accessories&Peripherals Cables&Accessories Cables USBCables | 943.601635   | 379.939371       | 59.515723           | 13972.773585 | 4.148428 |
| Electronics Headphones,Earbuds&Accessories Headphones In-Ear                      | 2998.529412  | 983.568627       | 58.627451           | 78673.568627 | 3.894118 |
| Electronics HomeTheater,TV&Video Accessories RemoteControls                       | 1170.448980  | 438.938776       | 59.510204           | 703.775510   | 3.800000 |
| Electronics HomeTheater,TV&Video Televisions SmartTelevisions                     | 40974.666667 | 25399.216667     | 38.133333           | 11446.116667 | 4.206667 |
| Electronics Mobiles&Accessories Smartphones&BasicMobiles Smartphones              | 20593.397059 | 15754.441176     | 23.294118           | 36665.720588 | 4.100000 |
| Electronics WearableTechnology SmartWatches                                       | 8664.854839  | 2363.435484      | 69.677419           | 19805.774194 | 4.008065 |

Smart televisions have the highest average price compared to all other categories while smart watches have the highest offered average discount.

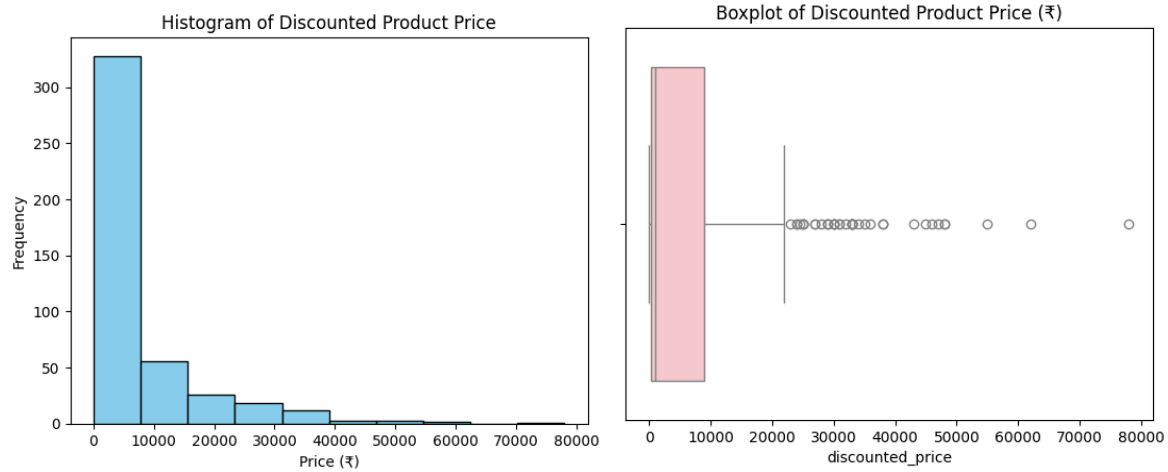
## Data Distributions

Figure 9: Distribution of Product Price



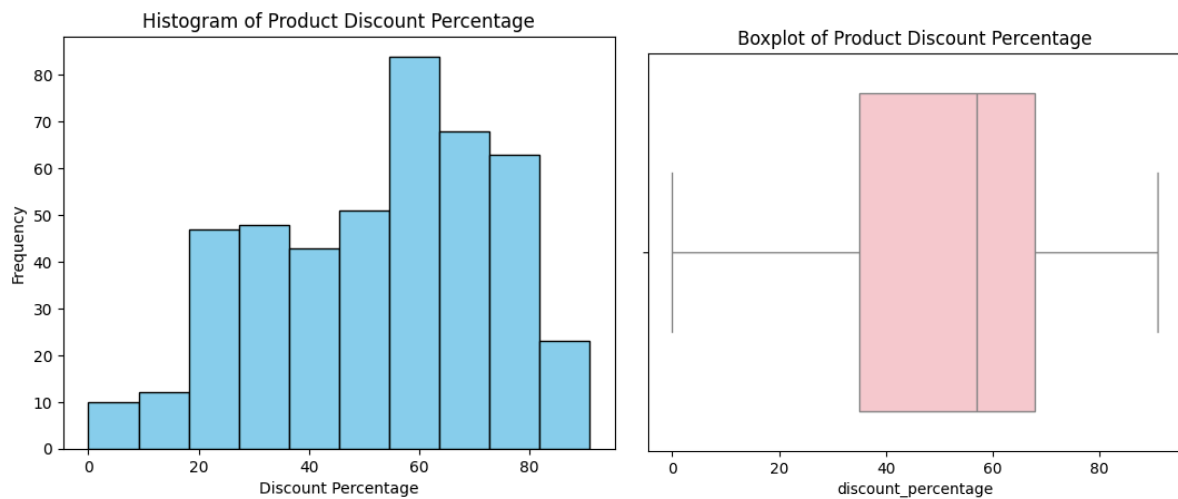
The product prices appear to be positively skewed due to many extreme data points.

Figure 10: Distribution of Discounted Product Price



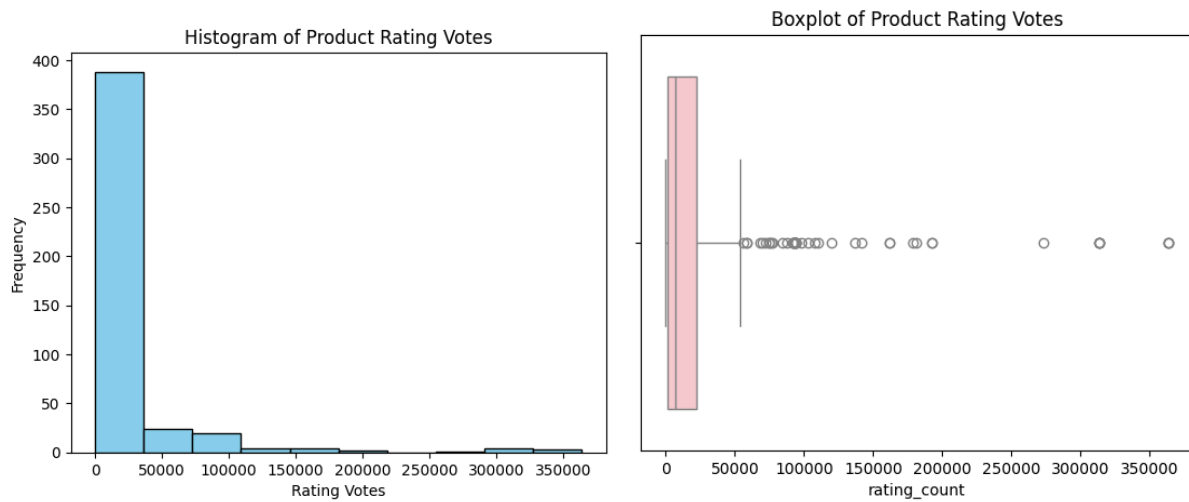
Similarly to the distribution of product price, the discounted price is also positively skewed.

Figure 11: Distribution of Product Discount Percentage



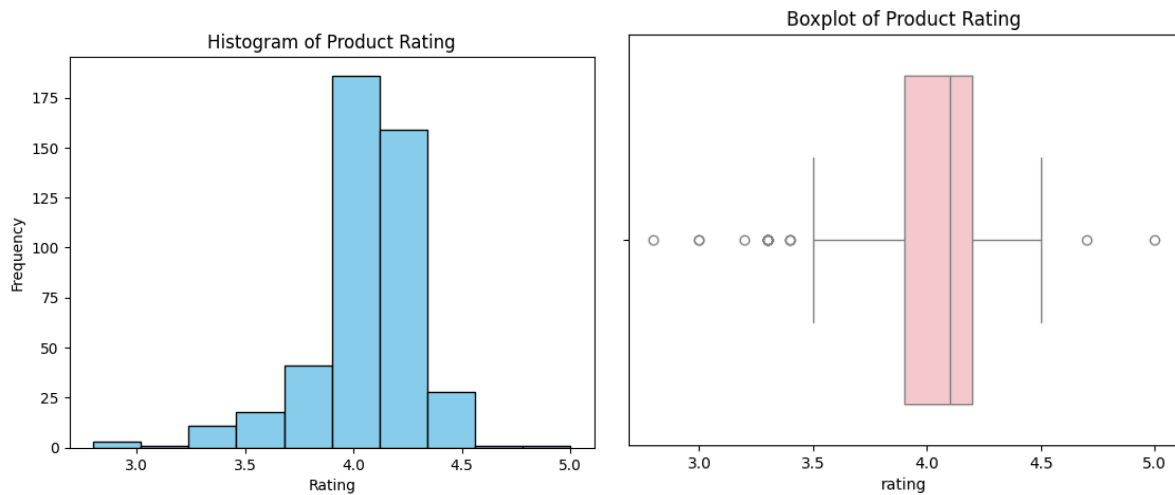
The distribution of the product discount percentage has a slight negative skew but is otherwise symmetric.

Figure 12: Distribution of Product Rating Votes



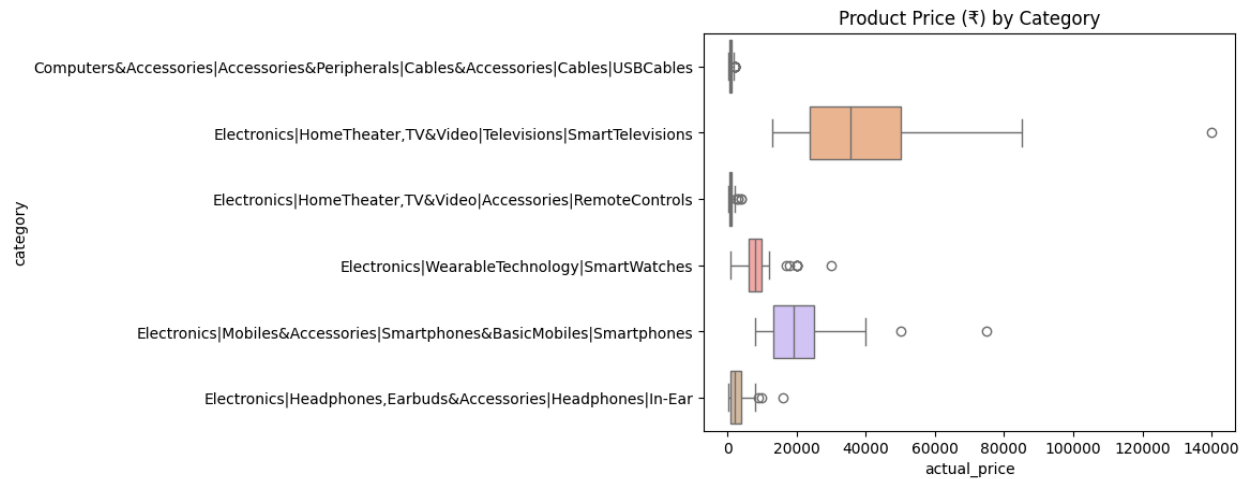
The distribution of the product rating votes has a strong positive skew, again indicating the presence of outliers.

Figure 13: Distribution of Product Rating



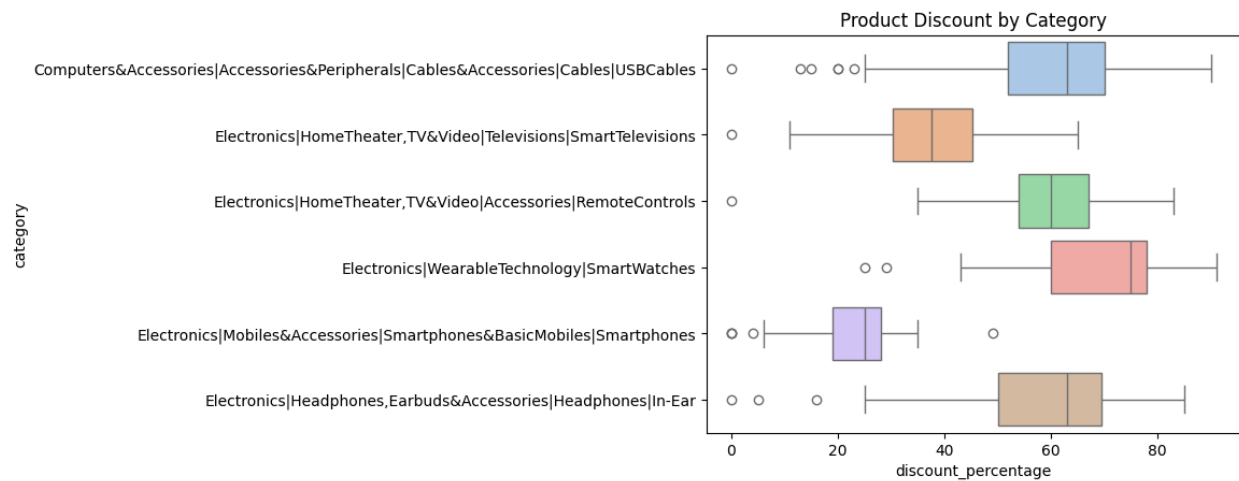
Product ratings appear to be uniformly distributed.

Figure 14: Distribution of Product Price by Product Category



The prices of smart televisions and smartphones appear to be nearly symmetrically distributed with the fewest outliers among all categories. These two categories also have the highest product prices.

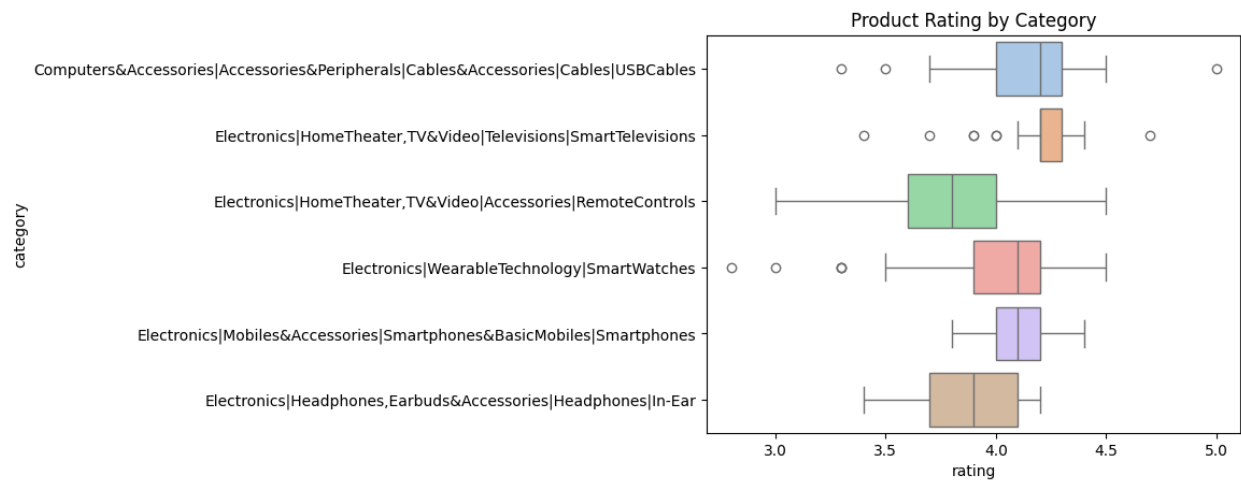
Figure 15: Distribution of Product Discount Percentage by Product Category



Smart watches have the highest offered discount while smartphones have the lowest.

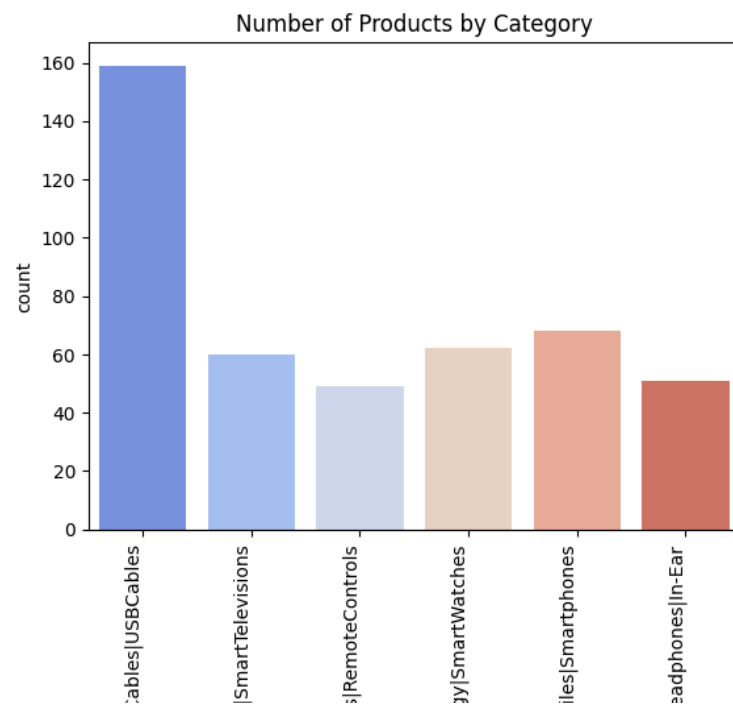
Figure 16: Distribution of Product Rating by Category





USB cables appear to be the highest rated product while remote controls are the lowest.

Figure 17: Distribution of Product Category



The majority of the products in the dataset are USB cables while there is little data on remote controls.

## Detection of Linear Relationships

Figure 18: Relationship Between Actual Product Price and Discounted Product Price



There appears to be a strong positive linear relationship between the actual and discounted product prices which would invalidate the assumption of multicollinearity if both variables are used as predictors in a regression model.

Figure 19: Relationship Between Discounted Product Price and Product Discount Percentage

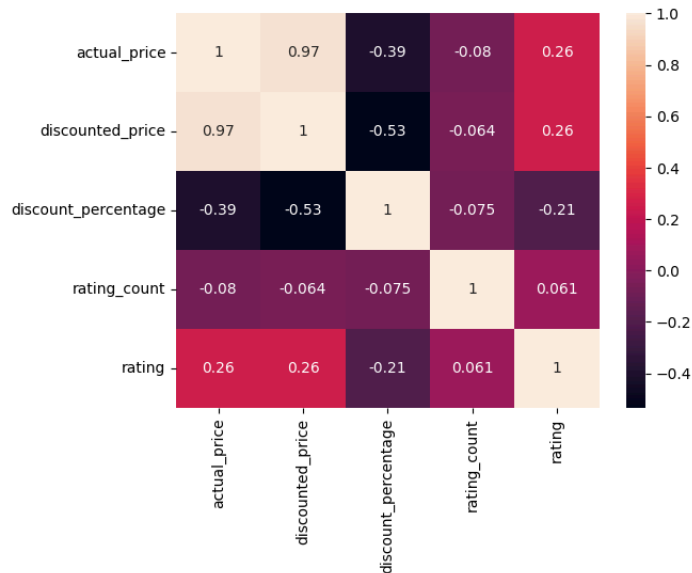


The plot suggests a potential moderate negative linear relationship between the discounted product prices and the discount being offered.

Figure 20: Correlation Matrix

|                     | actual_price | discounted_price | discount_percentage | rating_count | rating    |
|---------------------|--------------|------------------|---------------------|--------------|-----------|
| actual_price        | 1.000000     | 0.965297         | -0.393511           | -0.080022    | 0.259043  |
| discounted_price    | 0.965297     | 1.000000         | -0.534588           | -0.064209    | 0.262176  |
| discount_percentage | -0.393511    | -0.534588        | 1.000000            | -0.075121    | -0.206474 |
| rating_count        | -0.080022    | -0.064209        | -0.075121           | 1.000000     | 0.061368  |
| rating              | 0.259043     | 0.262176         | -0.206474           | 0.061368     | 1.000000  |

Figure 21: Correlation Heatmap



Based on *Figure 20* and *Figure 21*, prices and discounted product prices have a high correlation coefficient of about 0.97 indicating a strong potential linear relationship. A correlation of -0.39 suggests a moderate negative linear relationship between product prices and discounts while a correlation of -0.53 also implies the same between discounted prices and discounts offered.

## Data Preparation

The dataset has been prepared for the analysis by cleaning the numerical data of any unwanted characters and removing missing values. The number of products belonging to each category are highly imbalanced as several categories contain fewer than 30 products. For this reason and as previously discussed, the data has been filtered to include only those categories which have more than 30 unique products. As we are removing categories with very few observations,

the resulting dataset is still considered to be representative of the entire population of products within our data.

## **Data Transformation**

### **Classification**

The preliminary analysis uncovered three potential issues which need to be addressed prior to the model development phase: skewed distributions, outliers and multicollinearity. Decision trees for classification purposes are generally robust to outliers in contrast to KNN and logistic regression algorithms.

Due to the large volume and extremity of outliers, the treatment of outliers as missing values by applying imputation (TOMI) technique will be used. This method first requires the conversion of outliers to missing values, which are then imputed by the respective means for maximum machine learning efficiency (M. Seliem, 2022). Furthermore, a distance-based model like KNN is not only highly sensitive to outliers but also, skewed data. Min-max scaling will be applied to ensure all numerical features in the KNN model range between 0 and 1.

Z-score standardization (normalization) will be applied to all the numerical features to maintain interpretability and for fair comparison between regression coefficients. Multicollinearity in regression can increase the variance of the model coefficients which leads to a reduction in statistical power. We previously identified potential linear relationships of concern between some of the variables. However, this information alone is not sufficient for feature selection.

After constructing a baseline multinomial logistic regression model, the variance inflation factor (VIF) will be calculated for each feature to remove redundant variables. The odds ratio for each predictor will also be calculated to maximize the model's predictive power. Finally, if there appears to be no linearity between the log odds of the target and predictors, an appropriate transformation such as taking the logarithm or square root will be applied to conform to the model assumptions of logistic regression.

### **Information Retrieval**

For the vector space models, the product descriptions will undergo tokenization prior to the application of standard text preprocessing techniques which include: lowercasing, lemmatization, stopword removal, stemming and the removal of all nonalphabetic characters.

# Model Development

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## Model Inputs

All three of the baseline classification models will use the following features to predict the product category target variable.

- actual\_price
- discounted\_price
- discount\_percentage
- rating
- rating\_count

The vector space models will require only the *about\_product* feature and a new product description as inputs to output a ranked list of similar products from the data.

## Train-Test Split for Classification Models

The baseline models will be developed using an 80/20 stratified data split strategy which will reserve 80% of the data for model training and 20% of the data for testing purposes while maintaining class proportions. To prevent data leakage, the mathematical transformations discussed prior will be computed using only the training data.

Stratified k-fold cross-validation will be used to understand how the models generalize to unseen data and for hyperparameter tuning. After filtering the data, the remaining dataset retains only 30% of its original size. This shrink justifies the use of a 10-fold training strategy to prevent underfitting/overfitting while capturing sufficient product patterns.

## Logistic Regression

### Baseline Model

All of the numerical features have been normalized and the model has been trained and tested using an 80/20 split strategy.

### Verification of Assumptions

1. The product characteristics are not dependent on one another implying independent observations.
2. The product categories are mutually exclusive.
3. Analyzing the VIF for each variable confirms that either the product price or the discounted price should be removed in the final model to address the issue of multicollinearity as suggested during the preliminary analysis.

- The linearity between the log-odds of the target and the predictors can be verified using the Box-Tidwell test which can be done by adding the interactions between each predictor and its log-transformation to the model. The null-hypothesis of the test assumes that there is a linear relationship so interaction coefficients with p-values greater than the significance level of  $\alpha = 0.05$  are desired.

Figure 22: Box-Tidwell Test

|                        |          |                            |          |                               |          |
|------------------------|----------|----------------------------|----------|-------------------------------|----------|
| 0                      | 0.902643 | 0                          | 0.545898 | 0                             | 0.786968 |
| 1                      | 0.398525 | 1                          | 0.250998 | 1                             | 0.882458 |
| 2                      | 0.290396 | 2                          | 0.845499 | 2                             | 0.232719 |
| 3                      | 0.312356 | 3                          | 0.415203 | 3                             | 0.284362 |
| 4                      | 0.951060 | 4                          | 0.815493 | 4                             | 0.580592 |
| 5                      | 0.393115 | 5                          | 0.317644 | 5                             | 0.325268 |
| Name: actual_price_log |          | Name: discounted_price_log |          | Name: discount_percentage_log |          |
| 0                      | 0.874236 | 0                          | 0.109092 |                               |          |
| 1                      | 0.206716 | 1                          | 0.435323 |                               |          |
| 2                      | 0.926556 | 2                          | 0.767345 |                               |          |
| 3                      | 0.653377 | 3                          | 0.673897 |                               |          |
| 4                      | 0.419604 | 4                          | 0.905845 |                               |          |
| 5                      | 0.145834 | 5                          | 0.944123 |                               |          |
| Name: rating_count_log |          | Name: rating_log           |          |                               |          |

The p-values for all feature interaction terms for all category models are greater than  $\alpha$  which validates the assumption of linearity.

- Outliers have been imputed prior to building the model.

## Model Output

Figure 23: Multinomial Logistic Regression Model Coefficients

|                                                 | actual_price | discounted_price | discount_percentage | rating_count | rating    |
|-------------------------------------------------|--------------|------------------|---------------------|--------------|-----------|
| <b>Cables USBCables</b>                         | -3.778993    | -1.993861        | 0.542352            | -0.399315    | 0.823296  |
| <b>Headphones In-Ear</b>                        | -0.716802    | -0.338810        | 0.460671            | 2.167929     | -1.051029 |
| <b>Accessories RemoteControls</b>               | -1.900018    | -1.152788        | 0.303394            | -2.590959    | -0.595508 |
| <b>Televisions SmartTelevisions</b>             | 2.860498     | 1.875400         | 0.171509            | 0.018963     | 1.282237  |
| <b>Smartphones&amp;BasicMobiles Smartphones</b> | 1.336836     | 1.801814         | -2.663547           | 0.219221     | -0.293385 |
| <b>WearableTechnology SmartWatches</b>          | 2.198480     | -0.191755        | 1.185621            | 0.584161     | -0.165610 |

The implementation used to build the model does not assign a baseline category class which changes the interpretation of the model coefficients. Since the numerical features have been standardized, a one standard deviation increase in the product price increases the odds of the product belonging to the smart televisions category by a factor of  $e^{2.860498} \approx 17.470225$

compared to all other categories on average, assuming all other predictors are held constant. Positive coefficients of large magnitudes will lead to an increase in the odds while negative coefficients of large magnitudes will accomplish the opposite. For instance, product price is one of the most influential predictors of smart watches, USB cables and smart televisions.

## Model Evaluation

Figure 24: Multinomial Logistic Regression Classification Report

|                                      | precision | recall | f1-score | support |
|--------------------------------------|-----------|--------|----------|---------|
| Cables USBCables                     | 0.76      | 0.91   | 0.83     | 32      |
| Headphones In-Ear                    | 0.67      | 0.60   | 0.63     | 10      |
| Accessories RemoteControls           | 0.56      | 0.50   | 0.53     | 10      |
| Televisions SmartTelevisions         | 0.85      | 0.92   | 0.88     | 12      |
| Smartphones&BasicMobiles Smartphones | 0.92      | 0.86   | 0.89     | 14      |
| WearableTechnology SmartWatches      | 0.88      | 0.58   | 0.70     | 12      |
| accuracy                             |           |        | 0.78     | 90      |
| macro avg                            | 0.77      | 0.73   | 0.74     | 90      |
| weighted avg                         | 0.78      | 0.78   | 0.77     | 90      |

The model is able to predict the product category correctly 78% of the time. As seen by the *support* column, the class imbalance supports the use of the weighted average precision metric. A precision of 78% indicates that out of all the times the model predicted a product belonging to a particular category, 78% of those products were accurately classified. Similarly for recall, the model was able to identify 78% products actually belonging to a particular category. For instance, the model identified 58% of all smart watches. The majority of these scores are below 80% which suggests the need for model improvement.

## Final Model

The final model was built by excluding the discounted product price feature to prevent multicollinearity and was trained and tested using 10-fold stratified cross-validation.

## Model Evaluation

Figure 25: Multinomial Logistic Regression Final Scores

| Metric                          | Value |
|---------------------------------|-------|
| Average Accuracy                | 0.81  |
| Weighted Precision              | 0.83  |
| Weighted Recall                 | 0.81  |
| Average Training Time (Seconds) | 0.20  |
| Average Training Time (Seconds) | 0.03  |

The final scores were calculated by averaging the scores across all iterations of the cross-validation training strategy. The final model is reporting a higher accuracy, precision and recall in comparison to the baseline model which is expected due to the lack of multicollinearity.

## KNN

### Baseline Model

The numerical features have been scaled using min-max scaling and the model has been trained and tested using an 80/20 split strategy.

### Model Evaluation

Figure 26: KNN Classification Report

|                                      | precision | recall | f1-score | support |
|--------------------------------------|-----------|--------|----------|---------|
| Cables USBCables                     | 0.79      | 0.84   | 0.82     | 32      |
| Headphones In-Ear                    | 0.67      | 0.60   | 0.63     | 10      |
| Accessories RemoteControls           | 0.43      | 0.60   | 0.50     | 10      |
| Televisions SmartTelevisions         | 0.92      | 0.92   | 0.92     | 12      |
| Smartphones&BasicMobiles Smartphones | 0.93      | 0.93   | 0.93     | 14      |
| WearableTechnology Smartwatches      | 1.00      | 0.58   | 0.74     | 12      |
| accuracy                             |           |        | 0.78     | 90      |
| macro avg                            | 0.79      | 0.75   | 0.76     | 90      |
| weighted avg                         | 0.80      | 0.78   | 0.78     | 90      |

By default, the model predicts the target class using the 5 nearest neighbours. The model is able to correctly classify the product category 78% percent of the time. Out of all the category predictions made by the model, 80% of those predictions reflect the true product category. For recall, the model is able to identify 78% of the products belonging to a particular category. For instance, 84% of all USB cables are retrieved by the model.



## Final Model

The best value for the number of neighbours was determined to be 3 after performing hyperparameter turning and the final model was trained and tested using 10-fold stratified cross-validation.

## Model Evaluation

Figure 27: KNN Final Scores

| Metric                          | Value |
|---------------------------------|-------|
| Average Accuracy                | 0.81  |
| Weighted Precision              | 0.81  |
| Weighted Recall                 | 0.81  |
| Average Training Time (Seconds) | 0.14  |
| Average Training Time (Seconds) | 0.03  |

The final model is reporting a higher accuracy, precision and recall when compared to the baseline model which is likely a result of the hyperparameter tuning process. Notably, the final KNN model is also reporting a lower training time than the final logistic regression model.

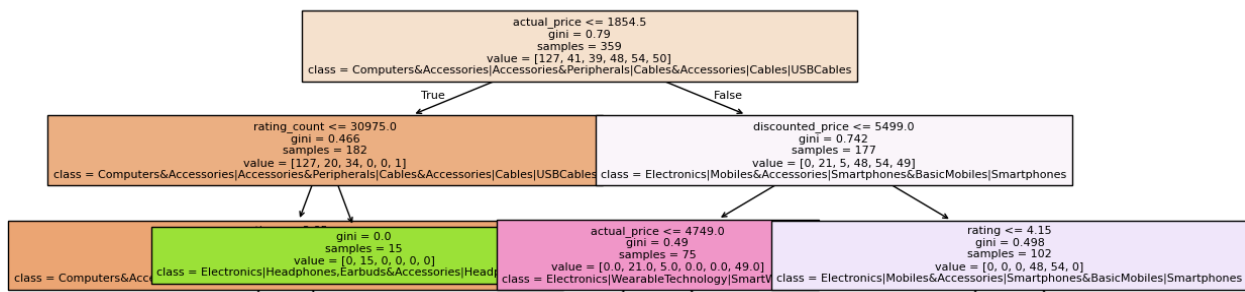
## Decision Tree

### Baseline Model

Decision trees are robust to scale differences and as such, the numerical features have not been transformed for this model.

### Model Output

Figure 28: Decision Tree Visualization



The product prices features appear to be the most important with the lowest Gini impurity when initially constructing the tree.

## Model Evaluation

Figure 29: Decision Tree Classification Report

|                                      | precision | recall | f1-score | support |
|--------------------------------------|-----------|--------|----------|---------|
| Cables USBCables                     | 0.78      | 0.91   | 0.84     | 32      |
| Headphones In-Ear                    | 0.82      | 0.90   | 0.86     | 10      |
| Accessories RemoteControls           | 0.50      | 0.40   | 0.44     | 10      |
| Televisions SmartTelevisions         | 0.85      | 0.92   | 0.88     | 12      |
| Smartphones&BasicMobiles Smartphones | 0.92      | 0.86   | 0.89     | 14      |
| WearableTechnology SmartWatches      | 1.00      | 0.67   | 0.80     | 12      |
| accuracy                             |           |        | 0.81     | 90      |
| macro avg                            | 0.81      | 0.77   | 0.79     | 90      |
| weighted avg                         | 0.81      | 0.81   | 0.81     | 90      |

The gini index is used as the split criterion and by default, the *max\_depth* parameter is set to 5 while the *min\_samples\_split* parameter is set to 2. The decision tree model has an ability to accurately classify the product category 81% of the time. From all the predictions the model makes for a particular class, 81% of the time the predictions correspond to the actual product category. For instance, when the model predicts a product to be a smart television, 85% of the time it is correct. In comparison, the model is able to identify 67% of all smart watches.

## Final Model

After hyperparameter tuning, the best value for the max depth of the tree was determined to be 7 while the minimum number of samples required within a node for further splitting was determined to be 5. The final model was trained and tested using 10-fold stratified cross-validation.

## Model Evaluation

Figure 30: Decision Tree Final Scores

| Metric                          | Value |
|---------------------------------|-------|
| Average Accuracy                | 0.90  |
| Weighted Precision              | 0.91  |
| Weighted Recall                 | 0.90  |
| Average Training Time (Seconds) | 0.08  |
| Average Training Time (Seconds) | 0.01  |

The final model is reporting significantly higher scores in comparison to the baseline model. It also reports the lowest training and testing times among the previously discussed models.

## Champion Model Selection

Figure 31: Performance Comparison for Classification Models

| Metric                          | Average Accuracy | Average Weighted Precision | Average Weighted Recall | Average Train Time (Seconds) | Average Test Time (Seconds) |
|---------------------------------|------------------|----------------------------|-------------------------|------------------------------|-----------------------------|
| Multinomial Logistic Regression | 0.81             | 0.83                       | 0.81                    | 0.33                         | 0.05                        |
| KNN                             | 0.81             | 0.81                       | 0.81                    | 0.14                         | 0.03                        |
| Decision Tree                   | 0.90             | 0.91                       | 0.90                    | 0.08                         | 0.01                        |

It is clear that the decision tree has outperformed both the KNN and logistic regression models across all performance metrics. The final model reports the highest accuracy, precision and recall with the lowest computing costs. The model accurately classifies a product's category 90% of the time. Anytime the model predicts a product to belong to a particular class, 91% of the prediction reflects the true product category. Finally the model is able to retrieve 90% of all products belonging to a particular category. The model is adept at minimizing both false positives and false negatives. For instance, the majority of products classified as a smart watch will actually be a smart watch and smart watches will rarely be classified as another type of product. The performance of this model can be attributed to the fact that decision trees do not require linear relationships within the data unlike most regression models. Such models are built using algorithms which are inherently robust to scale differences which is a key limitation of the distance-based algorithms used in KNN classifiers.

## Information Retrieval Pipeline

The product descriptions have undergone extensive text-preprocessing.

- Lowercase all letters
- Replace digits with the string *num*
- Tokenize the text
- Remove non-alphabetic tokens
- Remove stopwords
- Lemmatization
- Stemming

## Baseline Model

The TF-IDF statistical weight measure was used to identify relevant terms from the product descriptions in a unigram vector space model.

## Model Evaluation

Figure 32: Cosine Similarity Report

| Index | ID     | Category                                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cosine Similarity  |
|-------|--------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 0     | Doc301 | Electronics/Mobiles&Accessories/Smartphones&BasicMobiles/Smartphones | Processor: Octa-core MediaTek Helio G96 gaming processor clocked at 2.05GHz for excellent gaming performance along with a powerful Mali-G57 MC2 GPU  Display: 6.6 inch FHD+ (2412x1080) display comes with an ultra-smooth 120Hz high refresh rate display and a 90.8% Screen-to-body ratio 50MP + 2MP + 2MP AI triple rear camera setup with 50MP Mode, Ultra Steady Video, Night Mode, Panoramic view  Crystal clear 16MP Selfie Camera 5000mAh Massive Battery with 33W Dart charging support. The 33W Dart charger is included in the box with a USB Type-C cable  4 GB RAM LPDDR4X Fast memory   64 GB ROM UFS 2.1                                                                                                                                                                | 1.0000000000000002 |
| 1     | Doc396 | Electronics/Mobiles&Accessories/Smartphones&BasicMobiles/Smartphones | 4 GB RAM   64 GB ROM   Expandable Upto 512 GB  6.59 cm (6.53 inch) HD+ Display  13MP + 2MP + 2MP   5MP Front Camera  5000 mAh Lithium-Ion Polymer Battery  MediaTek Helio G35 Processor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.755078563597228  |
| 2     | Doc361 | Electronics/Mobiles&Accessories/Smartphones&BasicMobiles/Smartphones | Segment Best 16.21 centimeters (6.4-inch) Super AMOLED - Infinity U-cut display, FHD+ resolution with 90Hz Refresh rate, 800 Nits High Brightness Mode, protected by Gorilla Glass 5 Versatile 64MP+8MP+2MP+2MP Quad camera setup-64MP (F 1.8) main camera + 8MP (F2.2) Ultra wide camera+ 2MP (F2.4) depth camera + 2MP (2.4) Macro Camera  20MP (F2.2) front camera Monster 6000 mAh Battery   Memory, Storage & SIM: 4GB RAM   64GB internal memory expandable up to 1TB  SIM 1 + SIM 2 + MicroSD Android v12.0, One UI 4.1 operating system with MediaTek   Helio G80 Octa Core Processor 2GHz,1.8GHz 6000mAh lithium-ion battery, 1 year manufacturer warranty for device and 6 months manufacturer warranty for in-box accessories including batteries from the date of purchase | 0.7540448321927502 |
| 3     | Doc370 | Electronics/Mobiles&Accessories/Smartphones&BasicMobiles/Smartphones | Processor: MediaTek Dimensity 810 Octa-core 5G processor based on 6nm process with HyperEngine 2.0 and clock speed up to 2.4GHz Display: 6.6 inch FHD+ (2400x1080) Dot display with 90Hz high refresh rate and adaptive refresh rate technology, 240Hz touch sampling rate Camera: 50MP High resolution primary camera f/1.8 with 8MP Ultra-wide sensor  16 MP Front camera Battery: 5000mAh battery with 33W Pro fast charging support Memory, Storage : 6GB LPDDR4X RAM   128GB UFS2.2 Internal storage Headphones jack:3.5 mm.Display type:LCD.Operating system:MIUI 12.5 based on Android 11                                                                                                                                                                                       | 0.7506613073366025 |
| 4     | Doc385 | Electronics/Mobiles&Accessories/Smartphones&BasicMobiles/Smartphones | 4 GB RAM   64 GB ROM   Expandable Upto 512 GB 6.59 cm (6.53 inch) HD+ Display 13MP + 2MP + 2MP   5MP Front Camera 5000 mAh Lithium-Ion Polymer Battery MediaTek Helio G35 Processor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.7476442900047239 |

Figure 32 displays the top 5 products from the dataset which are the closest match to the product with an ID of *Doc301*. Cosine similarity is calculated by measuring the cosine of the angle between the search query vector and each of the vectors in the TF-IDF matrix. The similarity scores range between 0 and 1, with a higher score indicating a closer match. The model reports a perfect cosine similarity with *Doc301* as it is identical to the product of interest. *Doc396* is the next closest match with a cosine similarity score of about 0.76. Note that the top 5 product matches all belong to the smartphones category which suggests acceptable model performance.

## Final Model

The TF-IDF statistical weight measure was used to identify relevant terms from the product descriptions in a combined unigram and bigram vector space model.

## Model Evaluation

To understand how well the model performs across all product categories, the categories of the products retrieved were compared to the category of the product being searched by the model. The top 20 similar products identified by the model for every product in the dataset were used to calculate per-category retrieval accuracies.

Figure 33: Retrieval Accuracies by Product Category

| Category                                                                          | Average Accuracy |
|-----------------------------------------------------------------------------------|------------------|
| Computers&Accessories Accessories&Peripherals Cables&Accessories Cables USBCables | 0.93             |
| Electronics HomeTheater,TV&Video Televisions SmartTelevisions                     | 1.00             |
| Electronics HomeTheater,TV&Video Accessories RemoteControls                       | 0.85             |
| Electronics WearableTechnology SmartWatches                                       | 0.95             |
| Electronics Mobiles&Accessories Smartphones&BasicMobiles Smartphones              | 0.98             |
| Electronics Headphones,Earbuds&Accessories Headphones In-Ear                      | 0.75             |

Figure 33 implies that when the model is asked to find similar USB cables from the dataset, on average, 93% of all products retrieved are also USB cables. The average accuracies are all above 75% which again suggests acceptable model performance.

## Stakeholder Value

The technical analysis discussed prior will be translated into actionable insights to support decision making processes for the sales and marketing teams.

### Findings for Question 1

*How does the price, rating and discount compare for products belonging to different categories?*

#### Product Price

Smart televisions are the most expensive product with an average price of ₹40,974 followed by smart phones with an average price of ₹20,593. USB cables and remote controls are the cheapest sold products with average prices of ₹943 and ₹1,170 respectively. The sales team should prioritize selling smart televisions and smart watches to customers to drive profit. The marketing team should consider promoting inexpensive products like USB cables and remote controls as add-ons when customers are purchasing related products. A model which can be used to identify related products will be later discussed in detail.

#### Product Discount

Smart watches have an average offered discount of 70% which suggests that these products require greater attention from the marketing team to drive sales and ensure optimal usage of warehouse storage. Smart televisions and smartphones have average offered discounts of only 38% and 23% respectively. This indicates that these products are selling well and should continue to be the focus of sales employees when conversing with potential customers.

## Product Rating

USB cables, smartphones and smart televisions all have an average rating of about 4.1. It is not uncommon for customers to favour purchasing a product with a higher rating than another and for this reason the marketing teams should develop targeted promotion strategies to ensure these products are reaching the right customers. On the other hand, both remote controls and in-ear headphones have an average rating of about 3.8 which raises concerns for the sales teams to consider changes to future procurement strategies to instead supply products of better quality.

## Findings for Question 2

*How can the product characteristics be used to predict the product category?*

The decision tree model should be used to classify the category of a new product so that the most appropriate preexisting marketing content can be used for its promotion. This model outperformed all other models and will significantly reduce the risk of misclassifications of product categories. When this model is used to classify the category of a newly added product, 90% of the time its output will be correct.

Furthermore, the coefficients of the logistic regression model provided some insights into the most influential product characteristics when it comes to predicting product categories. These characteristics are summarized in the table below.

Figure 34: Key Characteristics by Product Category

| Category          | Influential Characteristics |
|-------------------|-----------------------------|
| USB cables        | Price                       |
| Remote controls   | Total ratings               |
| Smart televisions | Price                       |
| Smart phones      | Discount                    |
| Smart watches     | Price                       |

Based on *Figure 34*, the marketing team should ensure a reliable price is provided for a potential new smart watch product as it would be one of the most influential factors in determining its category if not previously known.

## Findings for Question 3

*How can product descriptions be used to identify similar products?*

The vector space model has been shown to be highly effective at using product descriptions to identify related products. For instance, if a sales employee is being asked by a customer to find products similar to a popular smartphone, using such a model will ensure that 98% of the top 20 matched products also belong to the smartphones category.

## Limitations and Ethical Implications

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### Limitations

- Prices are in Indian Rupees which reduce the interpretability for the North American analytics, marketing and sales teams.
- No information is provided regarding the purchase history for each product, making it difficult to identify which products require more attention from the marketing and sales departments.
- After filtering the dataset to include only relevant product categories, the remaining data entirely corresponds to products with various subcategories in the electronics department. This affects the ability to deliver insights for products belonging to other categories.

### Privacy & Ethics

- The dataset has a Creative Commons (CC) license which provides the necessary permissions to freely use the data for non-commercial purposes.
- The dataset includes the names and user IDs of individuals who contributed to product reviews. Sensitive information such as this does not concern the scope of this project and therefore efforts will be made to ensure this data does not get shared, modified or reproduced in any way.

## Technical and Theoretical Improvements

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- Consider adding a regularization penalty to the logistic regression models to better capture feature importance.
- Instead of solely using performance metrics such as accuracy, precision and recall for model evaluation, consider the use of statistical methods (likelihood-ratio test, Wald test, etc.) to evaluate the significance of the predictors and the model as a whole.
- Consider an exhaustive list of hyperparameter combinations for model tuning.

## External Files

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All work pertaining to this project can be accessed via the following GitHub repository:  
<https://github.com/harman-khehara/Amazon-Product-Analysis>.

# GenAI Disclosure

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The entirety of this project will be completed without the use of any generative artificial intelligence. Development history of the project will be tracked using GitHub and Google Suite. Justification of all work can be provided upon request.

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